

BAAZ JKBOSE



2026 CLASS-11TH

Body Fluids and Circulation

LECTURE-01

Zoology

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Topics To Be Covered

- 1 Blood.

Revision

- Breathing & Gaseous Exchange -

Disorder

- Occupational Lung Disease

↳ It occurs in persons working in Coal Mines, Grinding factories etc.

↳ Immune system/cells are unable to Break → leads to Difficulties in Breathing or remove Foreign substance.

↳ Leading to Damage, Scarring (Deposition of fibres).



(e.g) Silicosis.
Coal Worker Pneumoconiosis.

Body Fluids & Circulation.

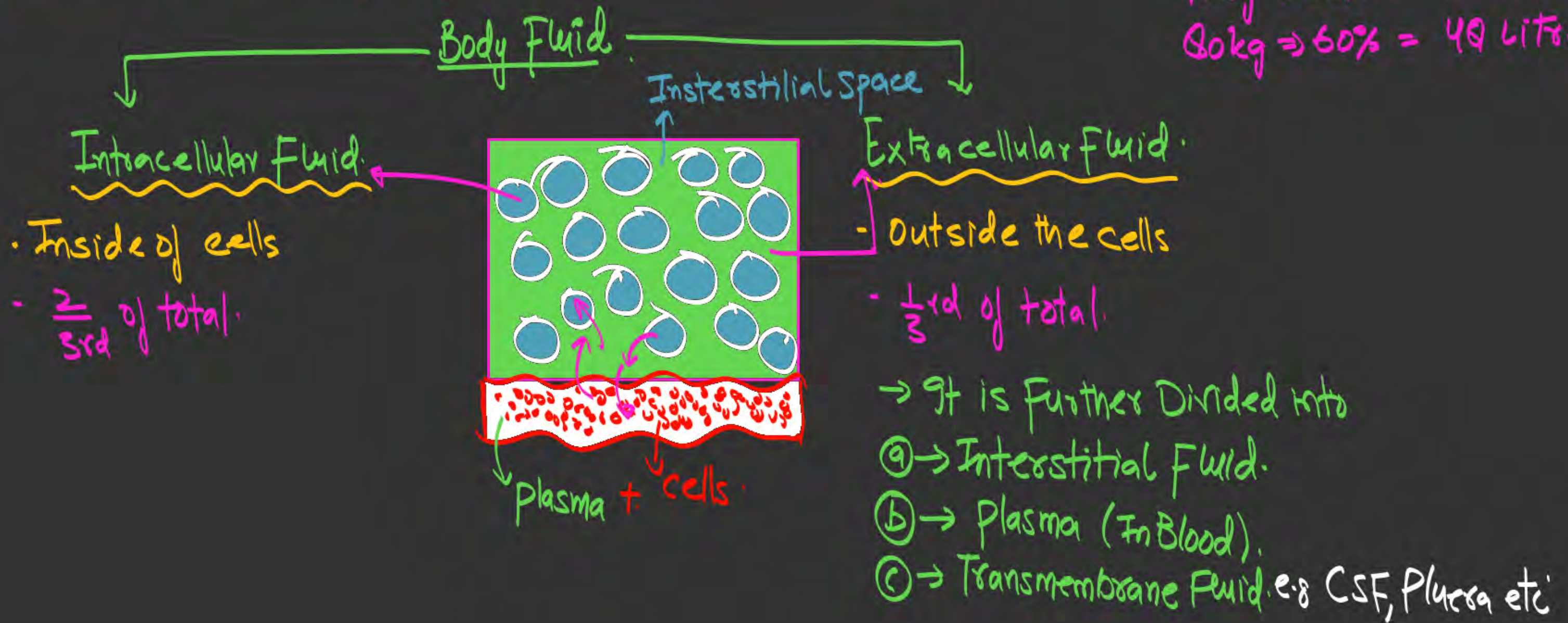
- Body Fluid - [Fluid. Tendency to flow, e.g. Liquid, Gas]
- We know, Human Body by weight is 60% Made of H_2O .

eg

$$60kg \Rightarrow 60\% = \underline{36L(H_2O)}$$

$$70kg \Rightarrow 60\% = 42 Litres$$

$$80kg \Rightarrow 60\% = 48 Litres$$



Blood

- It is our Main Circulatory Fluid.
- It is a type of Fluid Connective Tissue.
- It is made of: Plasma & Formed Elements.

① Plasma → It is Straw Coloured (pale yellow)

→ It is Liquid Content of Blood.

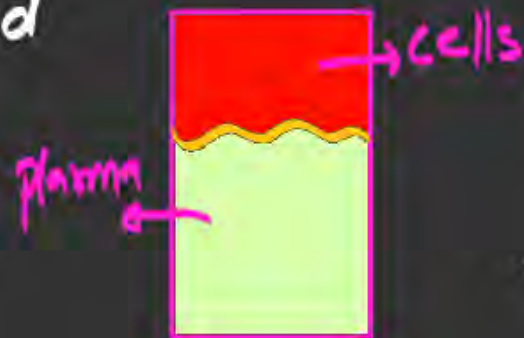
→ 55% of Blood.

→ It includes:

① H_2O (90-92%).

✓ ② Plasma proteins (7-8%).

③ Ions & Nutrients (1%).



② Formed Elements

→ It is cellular Content of Blood.

→ 45% of Blood.

→ It includes RBC, WBC & platelets.

Plasma proteins

- These are proteins present in plasma.
- They are of three basic types.

(a) Albumin

→ 60%

→ Globular proteins

→ Maintain Osmotic Balance



(b) Globulin

→ 35%

→ Globular protein

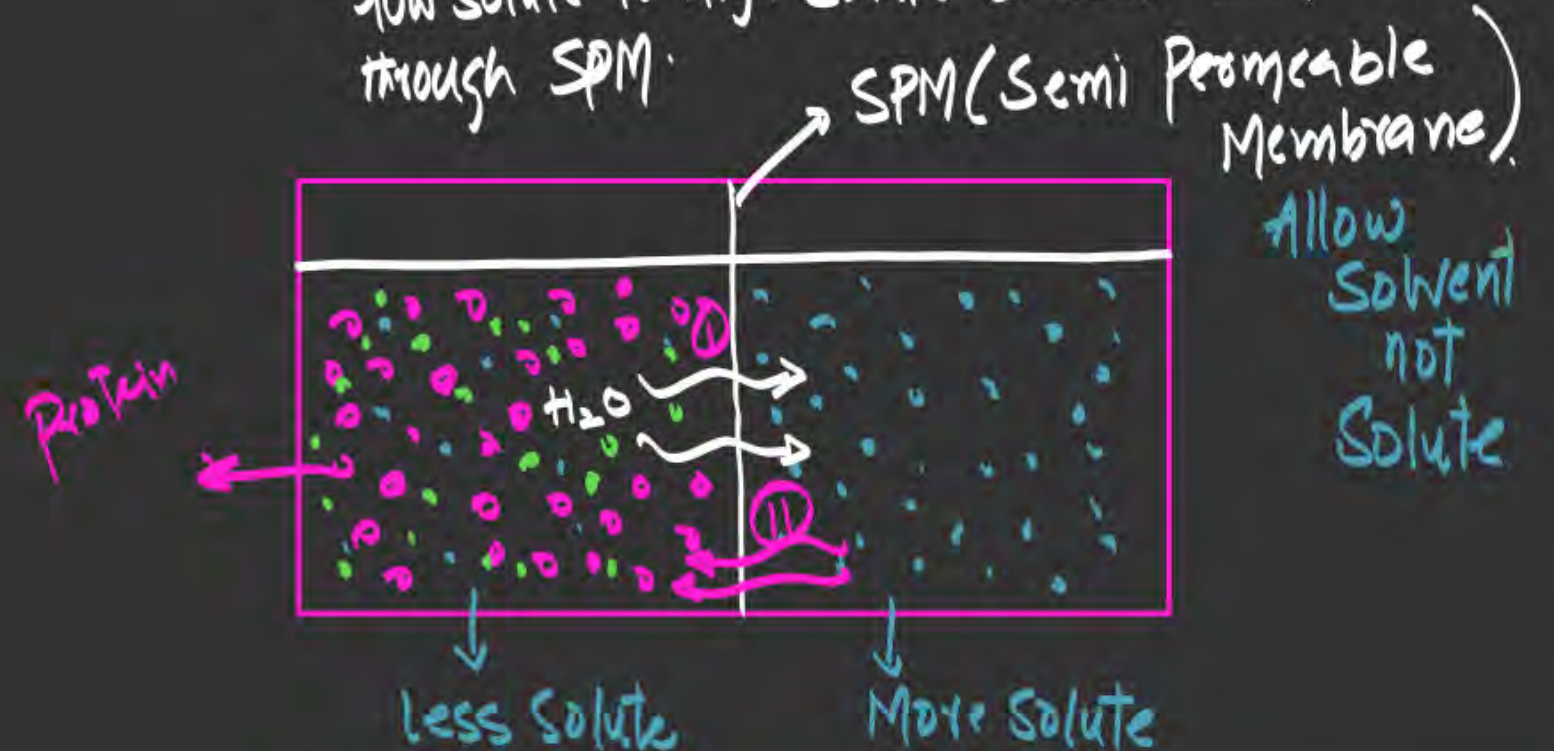
→ Help in Transport of substance, Defense Mechanism etc
(Antibodies)

(c) Fibrinogen

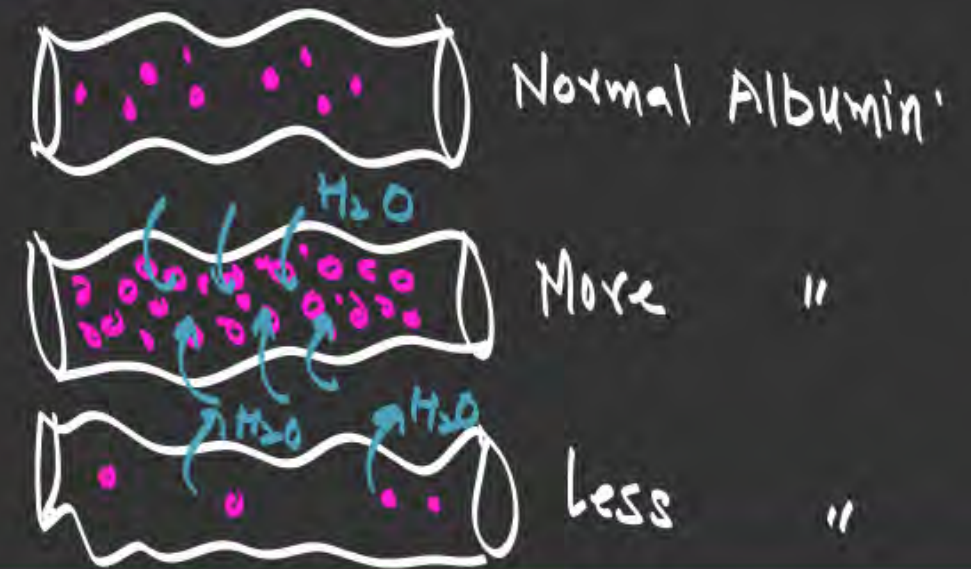
→ 5%

→ Help in clotting

Osmosis - Movement of Solvent from low Solute to High Solute concentration through SPM.



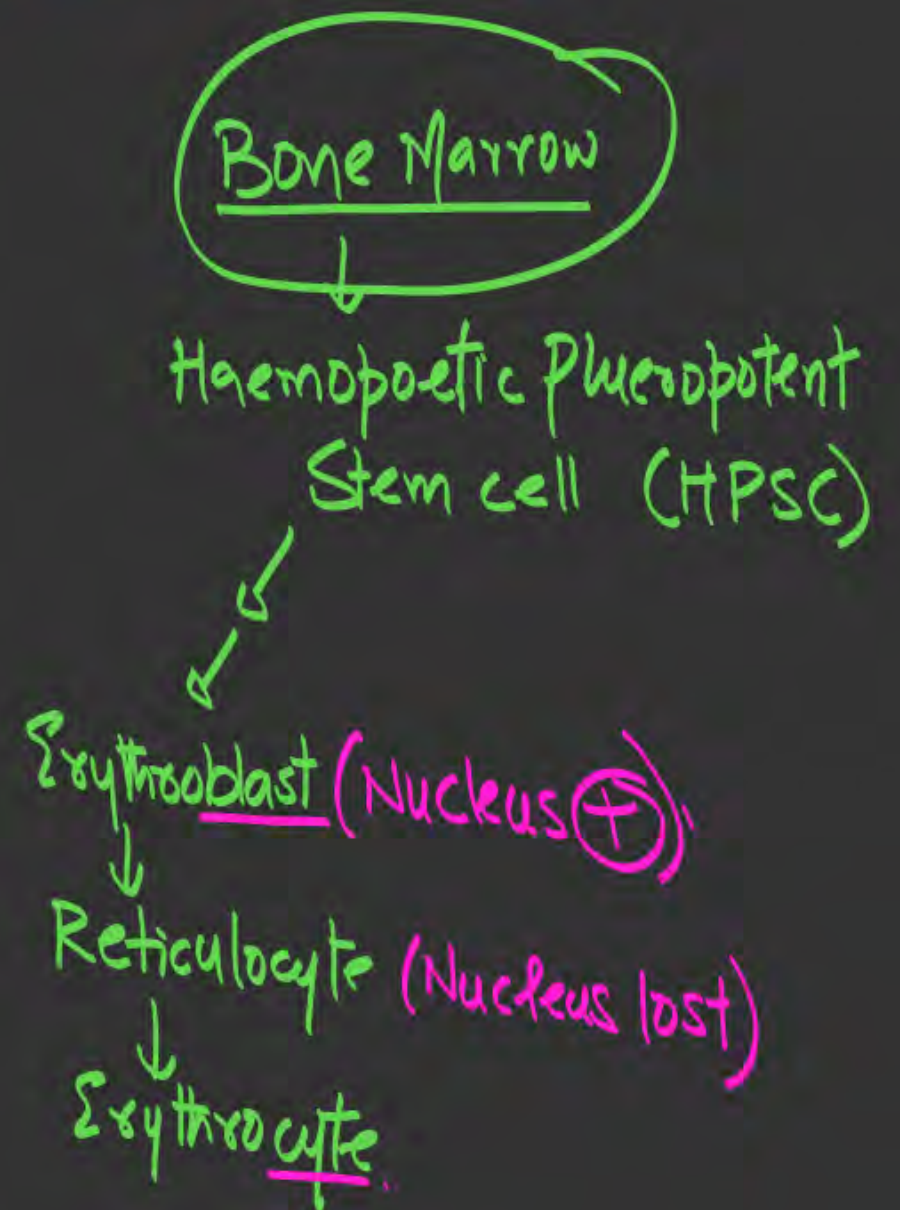
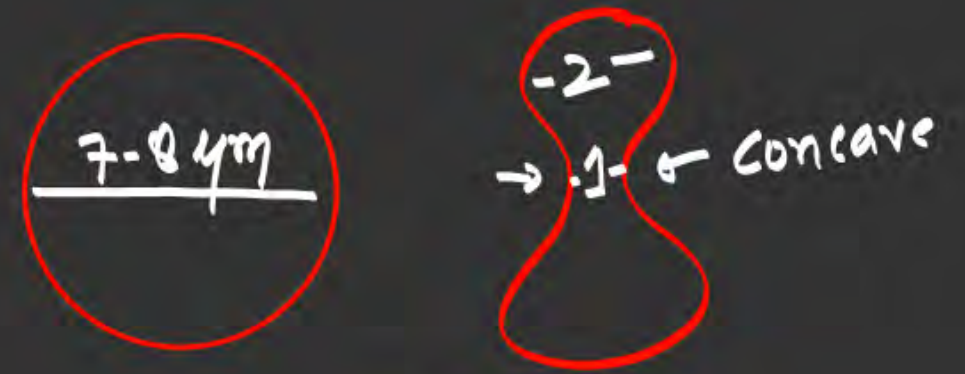
→ Albumin Function



⑥ Formed Elements

① RBC - (Red Blood Corpuscles)

- Mature RBC is called as Erythrocyte.
- Human Erythrocyte is Anucleated, Round, Biconcave.
- They lose Nucleus at Reticulocyte Stage.
- Majority of its space is filled with Haemoglobin.
- Haemoglobin is a protein pigment.
- Each Hb molecule is made of 4 Hb chains $[\alpha_2\beta_2]$.
- Each chain can hold one O_2 molecule.
- Normally an adult has 12-16 gm Hb/100ml of blood.



- RBC Formation occurs in Bone Marrow → Klas Erythropoiesis
- It is regulated by Hormone → Erythropoietin [From = Kidney]
- Number of RBC's = 5-5.5 Million / mm³

→ Decrease in No. of RBC = Erythrocytopenia.

→ Abnormally Rise in RBC's = polycythemia.

→ Anemia → Decreased oxygen carrying capacity of Blood.
→ Due to low RBC, low Hb or Both.
→ Cause - Fe deficiency

→ What is life span of RBC's 100-120 days.

→ After this old RBC's Break while passing Narrow Capillaries of Spleen.

→ Spleen is k/as graveyard of RBC.

(H/W) - Metabolism of Hb.

Thank
You